

# Adem Toumi

AI Engineer | Data Engineer | Data Analyst

+213 795 016 471 | Biskra, Algeria | a.toumi@esi-sba.dz | LinkedIn | GitHub

## PROFESSIONAL SUMMARY

AI, Data & Software Engineer graduating July 2026 from ESI-SBA with a dual State Engineer's Degree and M.Sc. in Artificial Intelligence & Data Science. Combines analytical depth with practical software engineering to build complete, production-ready intelligent systems — spanning real-time data pipelines, machine learning models, and scalable applications.

## EDUCATION

### Higher School of Computer Science (ESI-SBA)

State Engineer's Degree & M.Sc. in Artificial Intelligence and Data Science

Sidi Bel Abbès, Algeria

Sep. 2021 – Jul. 2026

- **Dual Thesis (Engineering & M.Sc.):** *Log-Based Anomaly Detection in Distributed Systems* — research and engineering work submitted for both the State Engineer's Degree and the Master's degree; benchmarked ML, deep learning, and transformer-based approaches across industrial log datasets and built a real-time streaming pipeline with cloud inference and an explainability dashboard.
- **Coursework:** Machine Learning · Deep Learning · NLP · Computer Vision · Distributed Systems · Big Data Engineering · Software Engineering.

## TECHNICAL SKILLS

|                                    |                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>      | Python · SQL · TypeScript · JavaScript · Dart · Java                                                                           |
| <b>Data Analysis &amp; EDA:</b>    | Data Collection · Data Cleaning · Preprocessing · Data Wrangling · EDA · Visualization · Pandas · NumPy · Matplotlib · Seaborn |
| <b>Data Engineering:</b>           | Apache Kafka · Apache Spark · PySpark · Hadoop · Apache Airflow · Streaming Ingestion Pipelines                                |
| <b>Machine Learning:</b>           | PyTorch · TensorFlow · Scikit-learn · XGBoost · Supervised & Unsupervised Learning · Model Evaluation                          |
| <b>Deep Learning &amp; Vision:</b> | Computer Vision · Object Detection & Tracking · Deep Learning · OpenCV · Edge Hardware                                         |
| <b>NLP &amp; Generative AI:</b>    | Natural Language Processing · Large Language Models · Retrieval-Augmented Generation · Semantic Search · Explainable AI (XAI)  |
| <b>Software &amp; Deployment:</b>  | PostgreSQL · Firebase · FastAPI · Node.js · React · Next.js · Flutter · Docker · Git · Linux                                   |

## RESEARCH & INDUSTRY EXPERIENCE

### Research Intern – Computer Vision & Embedded Systems

Feb. 2025 – Jun. 2025

Electronic Systems Research & Development Unit (URD Laboratory), Sidi Bel Abbès, Algeria

- Developed a real-time aerial surveillance system for UAVs — trained object detection models on aerial datasets (mAP@0.5  $\approx$  0.85), integrated dual tracking pipelines, and deployed the full inference stack on edge hardware (Raspberry Pi 5 + Intel Movidius NCS 2).
- Designed the complete hardware-software architecture: custom UDP video streaming to a Ground Control Station, automated gimbal control that keeps tracked targets centered, and an interactive operator interface with live telemetry.

### IT Intern – Database & Network Administration

Sep. 2024

Sonatrach, Direction Maintenance, Biskra, Algeria

- Optimized SQL queries, maintained database schemas, and automated backup scripts in an industrial environment.
- Supported network administration and system monitoring tasks.

## SELECTED PROJECTS

### Artificial Intelligence, Data Engineering & Analytics

#### Log-Based Anomaly Detection in Distributed Systems

Engineering & Master's Graduation Thesis

Python, PyTorch, Kafka, PostgreSQL, Streamlit

- Engineered a complete real-time anomaly detection pipeline for distributed infrastructure: log streams flow through Apache Kafka, ML and transformer-based models run cloud inference in parallel, results persist to a database, and a live dashboard surfaces alerts with per-model metrics.
- Integrated XAI (Explainable AI) using SHAP and LIME — token-level attributions are computed on demand and displayed inline in the dashboard, making every anomaly flag fully interpretable.

#### **KnowledgeHub AI – Enterprise RAG Platform**

*Python, FastAPI, React, LLMs, Docker*

- Built a production-ready document intelligence platform: organizations upload their knowledge base, ask questions in natural language, and receive grounded answers with exact source citations.
- Delivered the complete stack — hybrid search engine, FastAPI backend, React frontend, Docker Compose deployment, and a comprehensive automated test suite.

#### **Object Detection & Tracking for UAV Applications**

*Python, PyTorch, Computer Vision, Edge AI*

- End-to-end aerial surveillance system covering object detection, multi-target tracking, edge deployment, and a Ground Control Station interface — full detail in Research Experience above.

#### **Multi-Class Malicious URL Detection**

*Python, Scikit-learn, XGBoost, Data Analysis*

- Built a multi-class cybersecurity ML classifier extracting character n-grams and structural URL features across 651,191 URLs, delivering 96.8% Macro F1 accuracy.

#### **Paraphrase Detection via Siamese Attention**

*Python, PyTorch, NLP, Deep Learning*

- Designed a sentence-level semantic similarity model using a Siamese Bi-LSTM architecture with multi-head attention trained on PAWS and QQP datasets.

#### **HajjPortal – Trilingual Pilgrimage Management Platform**

*Next.js 14, TypeScript, Socket.io, Tailwind*

- Developed a full-stack, trilingual (Arabic, English, French) web platform with Next.js 14 App Router, dynamic RTL/LTR layout support, JWT authentication, and WebSockets.

#### **KidLink – Cross-Platform School Communication App**

*Flutter, Dart, Firebase Auth, FCM Push*

- Built a cross-platform mobile application in Flutter & Dart with role-based workflows, Firebase FCM push notifications, and Cloud Firestore offline persistence.